



APPLICATIONS

- Battery-powered devices
- High-efficiency SMPS
- Embedded computing
- Input filters

FEATURES

- Size 6mmx6mmx4mm
- Semi-Shielded Construction
- Low DCR
- Low Stray Field
- Max Operating Temp +125°C
- RoHS/REACH-Compliant, Halogen-Free

ELECTRICAL CHARACTERISTICS

Parameter			Value	Unit
Inductance ⁽¹⁾	L	±20%	6.8	μH
Resistance	R_{DC}	typ	33	mΩ
Resistance _{MAX}	$R_{DC\ MAX}$	max	40	mΩ
Rated Current ⁽²⁾	I_R	typ	4.4	A
Saturation Current _{25°C} ⁽³⁾	$I_{SAT\ 25°C}$	typ	4.1	A
Saturation Current _{100°C} ⁽⁴⁾	$I_{SAT\ 100°C}$	typ	3.7	A
Resonance Frequency	f_r	typ	25	MHz

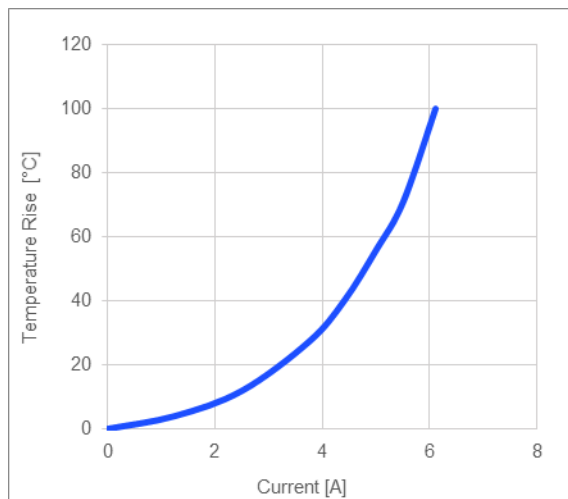
GENERAL SPECIFICATIONS

⁽¹⁾ Inductance	Measured at 100kHz, 100mA
⁽²⁾ Rated Current	Rated current will cause the coil temperature rise ΔT of 40K <i>I_R</i> measured with the inductor soldered in a single-layer PCB. Copper layer thickness 35μm Cu / PCB size 30x50mm. Temperature behavior dependent on circuit design, PCB layout, proximity to other components, and trace dimensions and thickness.
⁽³⁾ Saturation Current _{25°C}	Saturation current will cause L to drop from 30% at 25°C ambient temperature
⁽⁴⁾ Saturation Current _{100°C}	Saturation current will cause L to drop from 30% at 100°C ambient temperature
Temperature Test Condition	Electrical specifications measured at 25°C, 35% RH if not given differently
Operating Condition	Operating temperature: -40°C to +125°C (including temp rise) Should not exceed +125°C under worst-case operation conditions
Storage Condition	Tape and Reel packaging: -10°C to +40°C Humidity: <50% RH

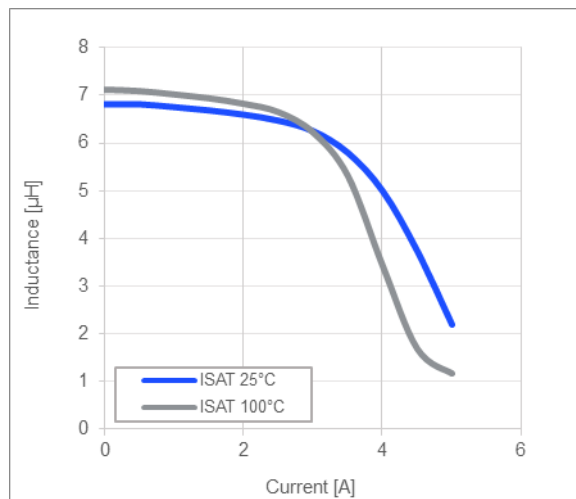
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TYPICAL PERFORMANCE CURVES

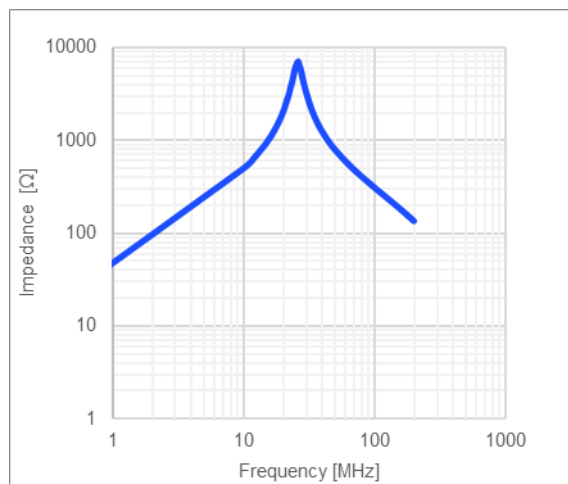
Temperature Rise vs. Current



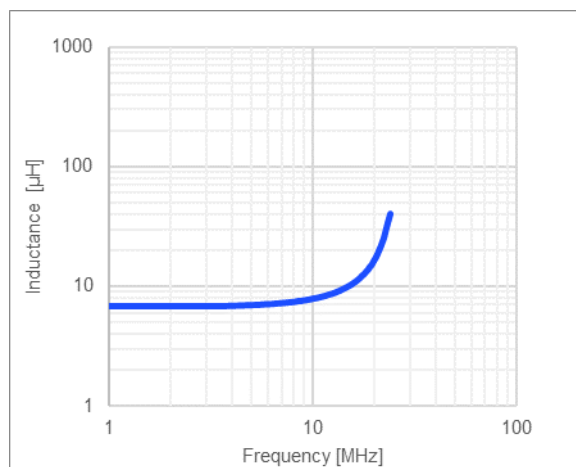
Inductance vs. Current



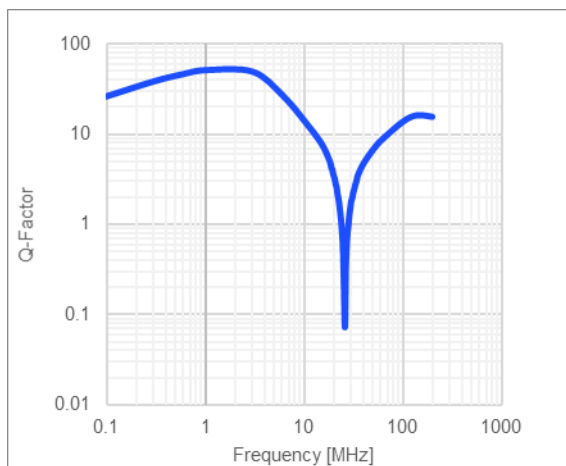
Impedance vs. Frequency



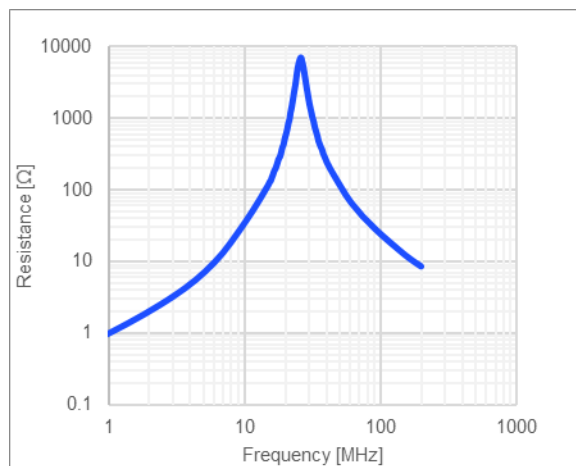
Inductance vs. Frequency



Quality Factor vs. Frequency



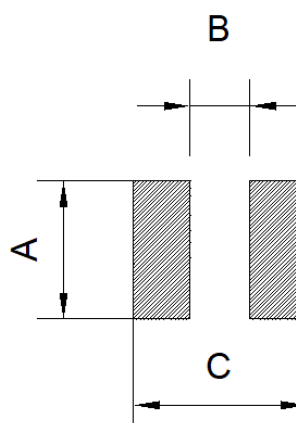
AC Resistance vs. Frequency



LAND PATTERN

Dimensions

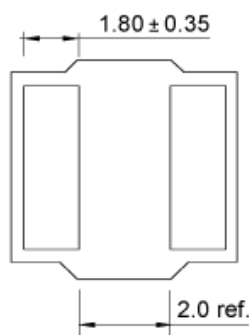
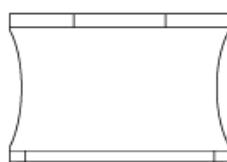
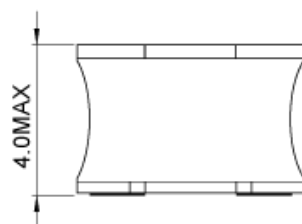
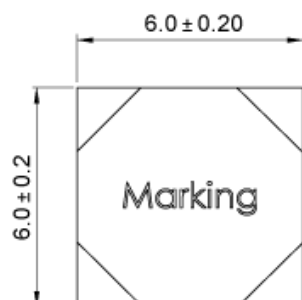
A	4.50 ref.
B	2.20 ref.
C	6.50 ref. (unit in mm)



PRODUCT PACKAGE AND DIMENSIONS

Dimensions

(unit in mm)



TOP MARKING

Marking

Inductance Code	6R8
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ORDERING INFORMATION

Part Number	$L^{(1)}$	R_{DC}	$I_R^{(2)}$	$I_{SAT\ 25^\circ C}^{(3)}$	$I_{SAT\ 100^\circ C}^{(4)}$
	typ (μH)	typ (mΩ)	typ (A)	typ (A)	typ (A)
MPL-SE6040-1R5	1.5	11.5	6.8	8.9	7.8
MPL-SE6040-2R2	2.2	14.5	6.3	7.2	6.7
MPL-SE6040-3R3	3.3	19.5	5.6	5.6	4.9
MPL-SE6040-4R7	4.7	23	5.2	5	4.5
MPL-SE6040-6R8	6.8	33	4.4	4.1	3.7
MPL-SE6040-8R2	8.2	39	4.0	3.6	3.2
MPL-SE6040-100	10	41	3.8	3.4	2.8
MPL-SE6040-150	15	70	2.8	2.7	2.4
MPL-SE6040-220	22	97	2.35	2.25	2

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(3) Saturation Current $_{25^\circ C}$	Saturation current will cause L to drop from 30% at 25°C ambient temperature
(4) Saturation Current $_{100^\circ C}$	Saturation current will cause L to drop from 30% at 100°C ambient temperature
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Storage Condition	Tape and Reel packaging: -10°C to +40°C Humidity: <50% RH

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